

RiskPulse

AI/NLP Financial Risk Engine for
Event-Driven Portfolio Stress Testing

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From unstructured financial signals to actionable portfolio risk

Problem & Approach

Turning Unstructured Financial Signals into Actionable Risk

⚠️ 01 — THE PROBLEM

Information Overload Continuous flow of unstructured news feeds and public market commentary.	Hidden Risk Signals Key sentiment, event classification, and severity buried in raw text.	Disconnected Decisions Event detection lacks direct linkage to portfolio stress loss testing.
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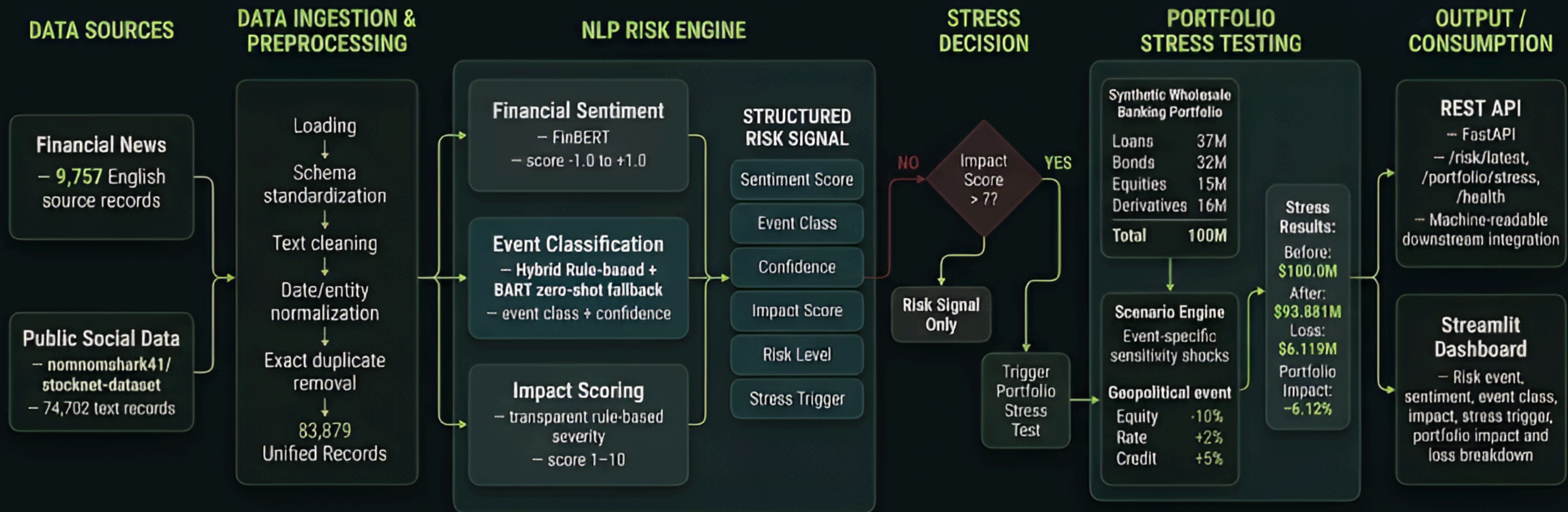
🔗 02 — OUR APPROACH (SOLUTION PIPELINE TABLE)

Unstructured Text → Structured Risk Vector → Quantified Loss

Stage	Data Input & NLP Engine	Extracted Parameters / Rule	System Output
① Ingest	Financial News & Social Media Continuous ingestion stream	Unstructured Data Stream	Raw Risk Signal
② Understand	AI / NLP Extraction Engine Extract contextual metrics	• Sentiment: -1 to +1 • Severity: 1 to 10 Scale	Structured Risk Vector
③ Trigger	Automated Threshold Rule Filter critical risk events	Impact Score > 7	Stress Test Triggered
④ Quantify	Scenario Stress Testing Engine Synthetic portfolio evaluation	\$100M Wholesale Portfolio	Quantified \$ Impact Loss

System Design

An end-to-end pipeline that converts financial text into structured risk signals and event-driven stress scenarios.



DEMONSTRATED EVENT: Boeing / BA → Iran War Strain → Sentiment +0.787 → Geopolitical → Confidence 0.95 → Impact 9.0/10 → Stress Trigger TRUE → Geopolitical Scenario → Portfolio Loss 6.12%

Implementation Highlights

Combining domain-specific NLP with transparent risk logic and an actionable stress-testing layer.

01 — FINANCIAL SENTIMENT

FinBERT · ProsusAI/finbert

Financial-domain model classifies Positive / Neutral / Negative and converts probabilities into a continuous -1 to +1 score.

Why: better fit for financial language. Output: sentiment_score

02 — EVENT CLASSIFICATION

Rules + BART zero-shot

Rule layer: high-confidence cues such as acquisition, sanctions, inflation, default, SEC. BART fallback: facebook/bart-large-mnli catches events outside predefined rules.

Output: event_class + confidence

03 — IMPACT SCORING

Transparent severity engine

Combines event baseline severity, sentiment intensity, and cues like default, bankruptcy, war, and sanctions into a reproducible 1–10 score.

Impact > 7 → Stress Test — transparent by design.

04 — PORTFOLIO STRESS ENGINE

Synthetic wholesale portfolio · \$100M

Loans, bonds, equities, and derivatives receive event-specific shocks through position sensitivities.

Example: Equity -10% · Rate +2% · Credit +5%

TECHNOLOGY STACK Python · Pandas · PyTorch · Hugging Face Transformers · FinBERT · BART · FastAPI · Streamlit

From Financial Signal to Portfolio Stress

The prototype converts unstructured text into an interpretable risk signal and quantifies its portfolio impact.

Risk Engine Validation

83.33% event-classification accuracy

- Evaluated on a curated set of 30 financial headlines
- 25 / 30 classifications were correct
- Hybrid approach combines high-confidence domain rules with BART zero-shot fallback

Demonstrated High-Impact Event

9.0 / 10 – Boeing / BA geopolitical event.

- Sentiment: +0.787
- Event: Geopolitical
- Classification confidence: 0.95
- Impact: 9.0 / 10
- Stress trigger: TRUE

Portfolio Stress Outcome

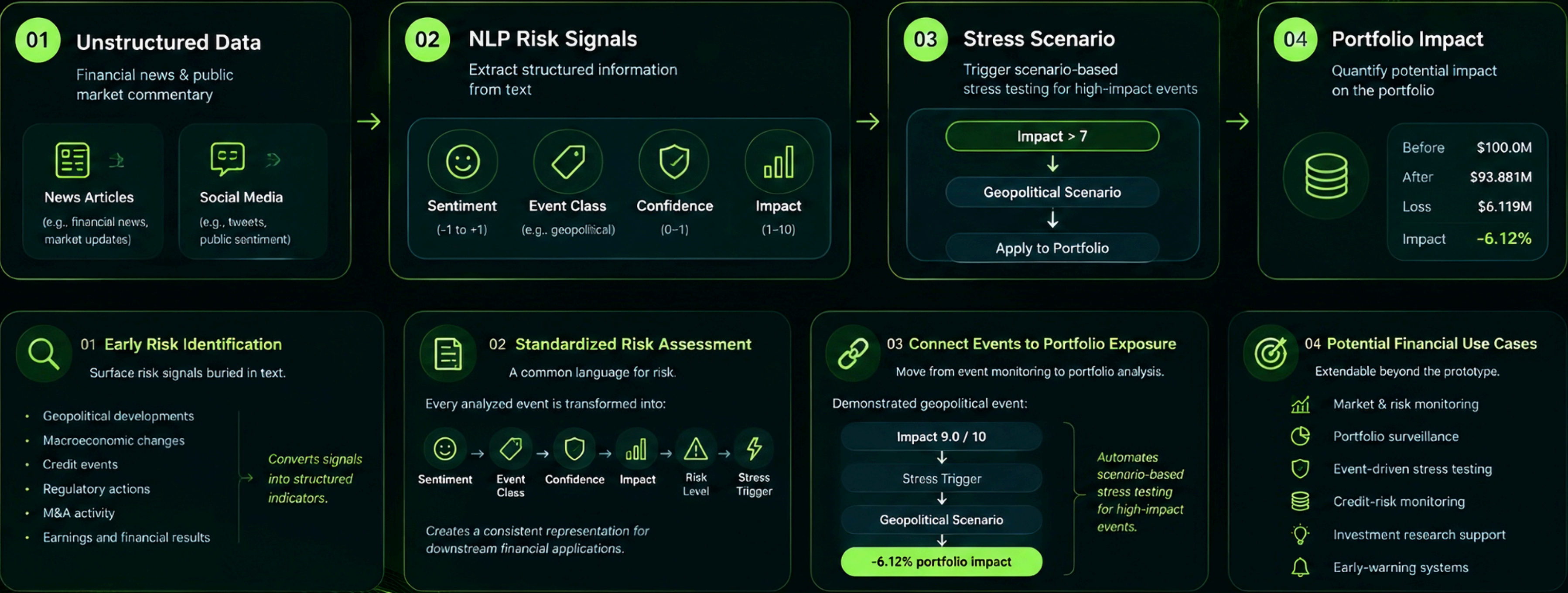
6.12% – Impact on the synthetic \$100M portfolio

- Portfolio before stress → \$100.0M
- Portfolio after stress → \$93.881M
- Portfolio loss → \$6.119M
- Portfolio impact → -6.12%
- Shocks: Equity -10% · Rates +2% · Credit +5%

The pipeline connects “what happened?” → “how severe is it?” → “what could it mean for the portfolio?”

Turning Unstructured Information into Actionable Risk

RiskPulse demonstrates how NLP-driven event detection can support faster, more structured financial risk assessment.



Current Boundaries & Next Steps

Current limitations

- Public historical datasets are used for reproducibility rather than a live streaming feed.
- Impact scores use a transparent rule-based severity framework, not a trained loss-prediction model.
- Portfolio shocks are scenario assumptions, not calibrated market forecasts or regulatory stress-test parameters.
- The \$100M portfolio is synthetic and does not represent an actual bank's positions or exposures.
- The 83.33% result comes from a curated 30-headline validation set and should not be generalized to all financial

Next Steps

1. Real-time ingestion
2. Learned impact prediction
3. Portfolio-aware stress testing
4. Model monitoring
5. Production integration

THANK YOU !